A local, non-spiking APL in the KC model

> **QUESTION**
>
> If APL is non-spiking, so that its inhibition stays near where it was excited, does it reshape the KC representation (the rank order of odor-pair similarity with inhibition on versus off) more like TNT vs TNT3c than one uniform APL or an APL weighted by per-KC synapse counts?

INPUTS

- The pinned KC model of Report 07 (KC input, odor gains, TNT and TNT3c targets)
- APL skeletons and geodesic distances (Report 10)
- Calyx KC ↔ APL and APL → PN synapses with their positions (Reports 01, 02)

OUTPUTS

- `odor_weighted/APL_local_model_summary.json`, `_fits.csv`, `_per_MB_{ORN,GH146}.csv`, `_odor_pairs.csv`, `_nulls.csv`
- `odor_weighted/APL_local_boutons_summary.json`, `_per_MB_{ORN,GH146}.csv`
- `odor_weighted/APL_local_model.png`, `APL_local_model_notes.md`
- the same in `with_outside_ROI_synapses/odor_weighted/`

SCRIPTS

- `scripts/apl_local_model.py` (inhibition on KCs)
- `scripts/apl_local_boutons.py` (inhibition on PN boutons too)
- `scripts/apl_local_model_figure.py`
- `scripts/reports/r11_prep.py` (matrix properties and checks for this report)
- `scripts/reports/r11_local_model.py` (this report)

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Reports 07 and 09 found that blocking APL in the data (TNT vs the TNT3c control) reorders odor pairs by similarity more than the model does: the rank correlation between blocked and intact similarity is 0.87 in the data (about 0.92 after correcting for recording noise) but 0.97–0.98 in the model with one uniform APL. That model treats APL as one unit with one activity per odor. APL is thought to be non-spiking, and its activity can stay local within its arbor (Amin et al. 2020, eLife). This report builds a version of the model in which inhibition spreads passively along APL's actual skeleton and fades with distance, and asks whether that reorders the pairs more. Hemibrain and male CNS only (3 mushroom bodies), both versions of the tables.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range

Symbol	What it stands for	Values, units, range
$k,\ j$	Kenyon cells (KCs); k receives inhibition, j excites APL	N = 1,793 (hemibrain R), 1,876 (male CNS L), 1,900 (male CNS R)
o	An odor	16 (ORN input) or 13 (GH146 input)
x_{ko}	Excitatory input of KC k to odor o (Report 07, pinned)	threshold units
y_{ko}	Response of KC k to odor o	≥ 0
θ	KC threshold	1
a	APL gain onto KCs, one per mushroom body and version, set so the mean intact fraction responding equals TNT3c's	> 0
M_{kj}	Coupling: how strongly activity of KC j inhibits KC k through APL	mean over k, j fixed (= that of C)
$u_k,\ v_j$	APL → KC synapses on KC k and KC → APL synapses from KC j in the calyx, each divided by its mean over KCs	mean 1
$\mathrm{in}(j),\ \mathrm{out}(k)$	APL's input sites from KC j (KC → APL synapses) and output sites onto KC k (APL → KC synapses), calyx only	skeleton nodes
$d(s,t)$	Geodesic distance between sites s and t along APL's skeleton (Report 10)	µm
λ	Length constant: distance over which a passive signal in APL falls to $1/e$ (37%)	2–200 µm, and ∞
c	Scale that sets the mean of the local M equal to the mean of C's	
$\ell_v,\ \kappa_v$	Length of the tree edge from node v to its parent, and its decay factor $\kappa_v = e^{-\ell_v/\lambda}$	µm; 0 to 1
B_{kb}	Share of KC k's claw PN synapses that come from bouton b	0 to 1, rows sum to 1
Q_{bo}	Local APL signal at bouton b for odor o	≥ 0
β	Gain of APL's divisive inhibition of boutons	> 0
f	Boutons' share of APL's calyx output synapses (Report 04)	0.097 hemibrain, 0.112 male CNS
S_{ab}	Cosine similarity of the KC population responses to odors a and b	0 to 1
ρ_{rank}	Spearman correlation over odor pairs between blocked and intact S_{ab}	−1 to 1; lower = more reordering
r	Pearson correlation over odor pairs between the model's and the data's ΔS_{ab} = blocked − intact	−1 to 1

1 What "non-spiking" means for the model

A spiking neuron turns local input into all-or-none action potentials that reach every part of the cell at full size, so "one activity per odor" is a fair description. A non-spiking neuron like APL instead carries graded voltage changes that spread passively and shrink with distance. In the simplest passive cable, a steady signal injected at one point falls off with distance d as $e^{-d/\lambda}$. The length constant λ of APL's fine calyx branches has not been measured, so the model scans it from 2 µm (very local) to 200 µm (almost global), and $\lambda = \infty$ gives back the one-unit APL.

Distances are along the skeleton, not straight lines. Report 10 showed that APL's calyx arbor comes in several sub-arbors joined farther down, so two points a few µm apart in space can be over 100 µm apart

along APL. For random pairs of a KC → APL site and an APL → KC site in the calyx the median distance is 193 µm (hemibrain R), 213 µm (male CNS L), 158 µm (male CNS R) (Figure 1a).

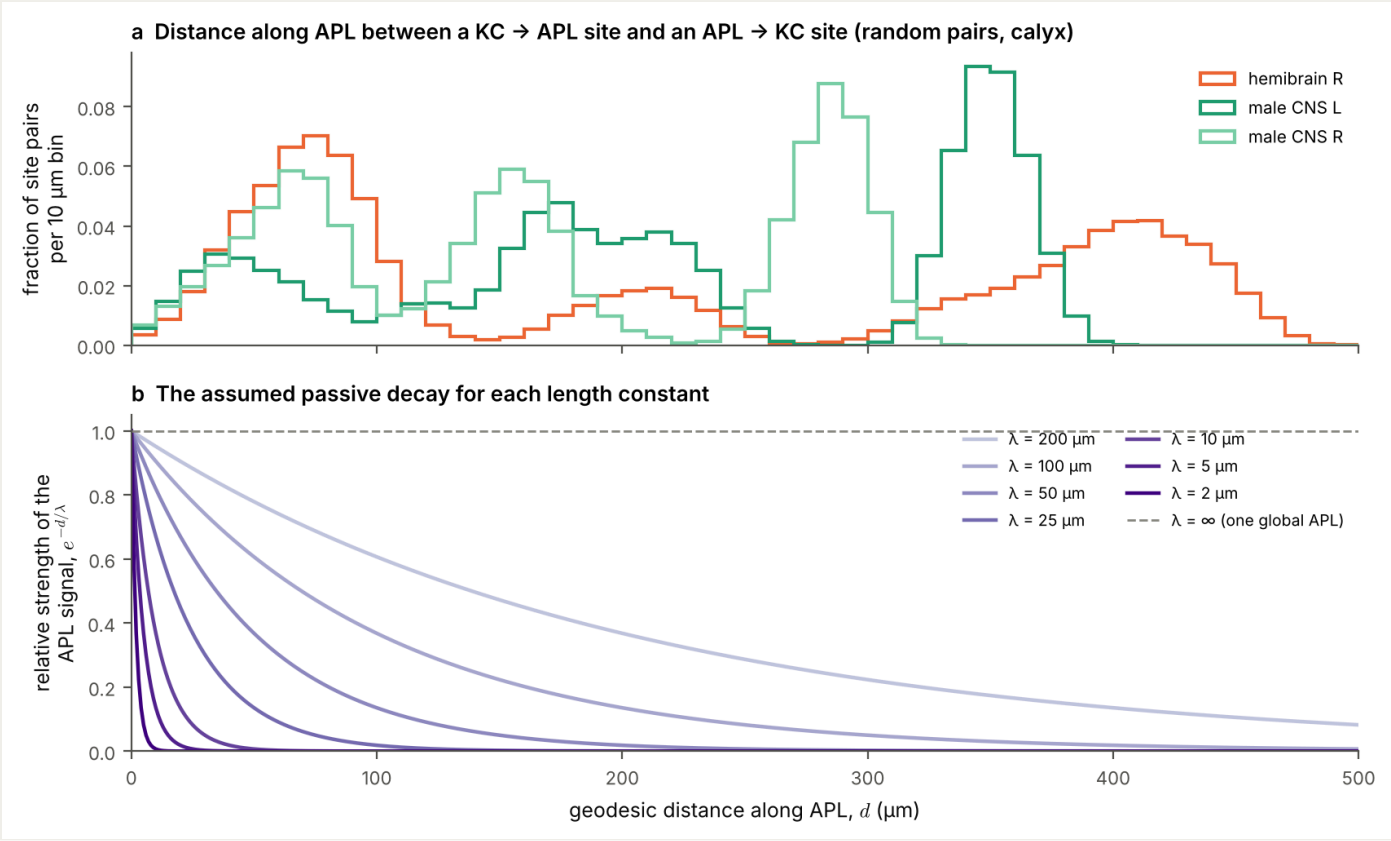


Figure 1. Distances inside APL and the decay that the local model assumes. a, for 300 random KC → APL sites and 300 random APL → KC sites in the calyx of each mushroom body (official ROIs), the distance between every input–output pair along the APL skeleton (each synapse placed at its nearest skeleton node). Lines: mushroom bodies; y-axis: fraction of the 90,000 pairs per 10 µm bin. **b**, the weight $e^{-d/\lambda}$ that a signal entering APL at one site has at another site d µm away along APL, for each length constant used. Shared x-axis: distance along APL in µm.

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2 The model, step by step

2.1 Step 1: KC input and pinning (unchanged from Report 07)

Each KC's input is its claws' PN synapse shares times the glomeruli's ORN (or GH146) responses, with negative responses set to 0. Each odor's input is scaled by a gain g_o so that, with APL blocked and $\theta = 1$, the fraction of KCs responding equals that odor's TNT fraction. Report 07 explains this pinning in full; it is the same here, so all versions start from identical blocked responses $y_{ko}^{\text{blocked}} = [x_{ko} - 1]_+$.

2.2 Step 2: feedback through a coupling matrix

With APL intact, each KC is inhibited by an amount that depends on the activity of all KCs, through the coupling matrix M:

$$y_{ko} = \left[x_{ko} - \theta - a \sum_j M_{kj}\, y_{jo} \right]_+ \tag{1}$$

Solved for each odor separately; the responses appear on both sides, so the solution is the self-consistent steady state.

Every version has this form and differs only in M. The versions:

- **U, uniform**: $M_{kj} = 1/N$. APL sums all KC activity and inhibits every KC equally. This is version U of Reports 05 and 07.
- **C, per-KC counts**: $M_{kj} = u_k v_j / N$. KC j excites APL in proportion to its KC → APL synapses, and KC k is inhibited in proportion to its APL → KC synapses, but APL is still one unit with one activity.
- **local**, length constant λ: Equation 2.

$$M_{kj} \;=\; c \sum_{s \,\in\, \mathrm{out}(k)} \; \sum_{t \,\in\, \mathrm{in}(j)} e^{-d(s,t)/\lambda} \tag{2}$$

KC j's activity enters APL at each of its KC → APL synapses t, spreads along APL with decay $e^{-d/\lambda}$, and inhibits KC k at each APL → KC synapse s on k.

As $\lambda \to \infty$ every exponential becomes 1, and $M_{kj} = c\,|\mathrm{out}(k)|\,|\mathrm{in}(j)|$, which is C (checked: correlation 1.000 in all three mushroom bodies). So C is the local model with an infinitely long length constant, and the local versions differ from C only in where on APL each KC's synapses are. The constant c sets the mean of M to that of C, so that the gain a has comparable values across versions; the calibration in Step 4 would absorb any constant anyway.

What counts as a site

Only calyx synapses are used, as for C (per-KC counts are calyx only, Yang 2026-10-21). APL's side of each synapse is placed at the nearest node of the APL skeleton: the T-bar for APL → KC, the PSD for KC → APL (median snap distance 0.4 µm in hemibrain R, 0.8 µm in male CNS L, 0.8 µm in male CNS R).

Two sites at the same node have $d = 0$. Snap distances are not added to d, so at λ = 2 µm the result depends on the skeleton's resolution.

2.3 Step 3: computing M on the tree

A direct sum over all pairs of sites (about 20,000 × 25,000 in hemibrain) would need every pairwise distance. Because a skeleton is a tree, the same sums can be computed exactly in two passes. First the tree is reduced to the part that joins the synapse nodes, with chains of unbranched nodes merged into single edges of summed length ℓ_v (16,471 nodes remain in hemibrain R, 4,513 and 5,082 in male CNS L and R). Let n_{vj} be the number of KC → APL sites of KC j at node v, and $\kappa_v = e^{-\ell_v/\lambda}$. Then:

$$\mathrm{up}_{vj} = n_{vj} + \sum_{w \,\in\, \mathrm{children}(v)} \kappa_w \,\mathrm{up}_{wj}, \qquad \Phi_{vj} = \mathrm{up}_{vj}\,(1 - \kappa_v^2) + \kappa_v\,\Phi_{\mathrm{parent}(v),\,j}, \quad \Phi_{\mathrm{root},j} = \mathrm{up}_{\mathrm{root},j} \tag{3}$$

Pass 1 runs from the leaves up; pass 2 from the root down. $\Phi_{vj} = \sum_{t \in \mathrm{in}(j)} e^{-d(v,t)/\lambda}$, the signal from KC j at node v.

Why it works: the path from v to a site t below v stays below v, and its weight is the product of the κ of its edges, which pass 1 collects. A site not below v is reached through v's parent; pass 2 takes the parent's total, multiplies by κ_v for the edge, and removes the part that came from below v itself (that part crossed the edge twice, hence κ_v^2). Finally $M_{kj} = c \sum_{s \in \mathrm{out}(k)} \Phi_{sj}$. **Check:** at λ = 10 µm, Φ for 40 random KCs at 60 random nodes was also computed by brute force (shortest paths on the tree); the largest difference relative to the largest value was 2e-16, 3e-16, 3e-16 (rounding error).

2.4 Step 4: one APL gain per version, and the solver

For each mushroom body, input table and version, the gain a is found by bisection (32 halvings after doubling until it is large enough) so that the mean fraction of KCs responding over odors equals TNT3c's, 6.83% for the 16 odors. Equation 1 is solved per odor with an exact **active-set** method: guess which KCs

respond, solve the linear equations $(I + aM_{SS})\, y_S = x_S - \theta$ for that set S, add KCs whose net input is still above threshold and drop KCs whose solution is negative, and repeat until the set stops changing. If the set cycles or the result fails a residual check, a damped fixed-point iteration is used instead. Both solvers were run on the same cases and agree (gains equal to 4 decimals, responses within about 10^{-6}). With feedback this strong a threshold-linear network could in principle have more than one steady state; the solver starts from the blocked responses and finds one.

2.5 Step 5: readouts

- **Rank order** ρ_{rank}: for each mushroom body, the Spearman correlation over odor pairs between blocked and intact cosine similarity, then the mean over the three mushroom bodies. Data: TNT vs TNT3c, 0.869 over the 16 odors, 0.873 over the 13 GH146 odors; corrected for recording noise about 0.92 (Report 09).
- **Pattern** r: the model's change in similarity per pair (blocked − intact, similarity averaged over the three mushroom bodies) correlated with the data's change (TNT − TNT3c). The data's own reliability limits this to about 0.84 (Report 07).
- **Size**: the mean change in similarity over pairs (data: 0.178 for the 16 odors).

2.6 Step 6: a null for the real placement of each KC on APL

Is it the real position of each KC's synapses on APL that matters, or only how local the inhibition is? For the local version with the best pattern r, KCs are shuffled within their class (αβ, α′β′, γ): KC k gets the APL sites of KC $\pi(k)$, which turns M into $M_{\pi(k)\pi(j)}$. This keeps the arbor, the class composition and the locality, and only breaks which KC sits where. 100 shuffles per mushroom body, each recalibrated. One-sided p = (1 + shuffles at least as extreme) / 101.

> **Caveat on the null**
>
> The λ tested is the one with the best r among the local versions, picked before the shuffles, so the p value does not correct for having tried seven λ values.

2.7 Step 7: local inhibition of PN boutons too

APL also inhibits the PN boutons (about 10% of its calyx output synapses, Report 04). The second script adds this route. Bouton b receives the local APL signal at its APL → PN synapses, per PN → KC synapse it makes onto claws, and divides its output:

$$x_{ko} = \sum_b B_{kb}\, \frac{g_o\, r_{g(b),o}}{1 + \beta\, Q_{bo}}, \qquad Q_{bo} = \frac{1}{n_b^{\mathrm{claw}}} \sum_{s \,\in\, \mathrm{APL}\to\mathrm{PN}(b)} \sum_j \Phi_{sj}\, y_{jo} \tag{4}$$

$g(b)$: the glomerulus of bouton b; n_b^{claw}: PN synapses on the claws whose main bouton is b. With $\beta = 0$ this is exactly the input of Step 1 (checked).

- **boutons only**: all inhibition on boutons ($a = 0$), β calibrated to TNT3c's fraction. An upper bound for this route.
- **KC + boutons**: KC gain $(1 - f)\, a_\lambda$ and bouton gain $f\, \beta_\lambda$ (each from its own calibration at the same λ), then both scaled together to TNT3c's fraction. This is the bouton route at its real synapse share.
- λ = ∞, 50, 25, 10, 5 µm.

3 What shorter length constants do to the coupling

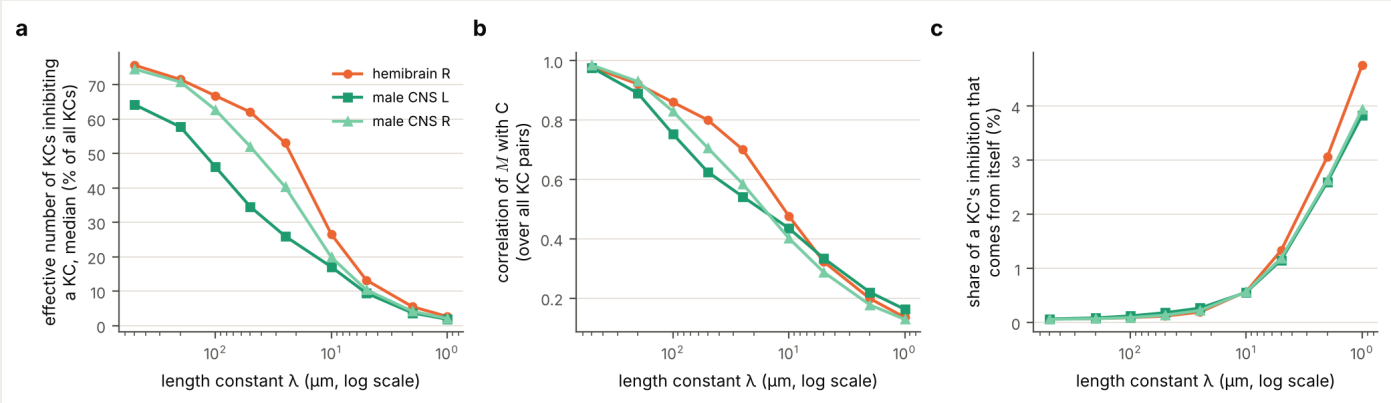


Figure 2. Shorter length constants make each KC's inhibition depend on fewer KCs. Properties of the coupling matrix M of the local model, per mushroom body (lines), as λ shrinks (x-axis, log scale, long λ on the left). **a**, for each KC k the participation ratio $(\sum_j M_{kj})^2 / \sum_j M_{kj}^2$, the number of KCs that would contribute equally to its inhibition, as a percentage of all KCs; median over KCs. With U it would be 100%. **b**, Pearson correlation between M and the C matrix over KC pairs (C = λ = ∞). **c**, mean over KCs of $M_{kk} / \sum_j M_{kj}$: the share of inhibition a KC would get from its own activity if all KCs were equally active. From `reports/data/r11_M_properties.csv`.

At λ = 100 µm each hemibrain KC is still inhibited by about 67% of all KCs, at 10 µm by about 27% and at 2 µm by about 5.5% (about 99 KCs). Self-inhibition stays small even at 2 µm (3.1%), because a KC's own APL synapses are only a few of the many near any site. Short λ also shifts the coupling toward KCs of the same class (42% of M at λ = ∞, 62% at 2 µm in hemibrain), since KCs of one class share parts of the calyx.

4 Results: rank order and pattern

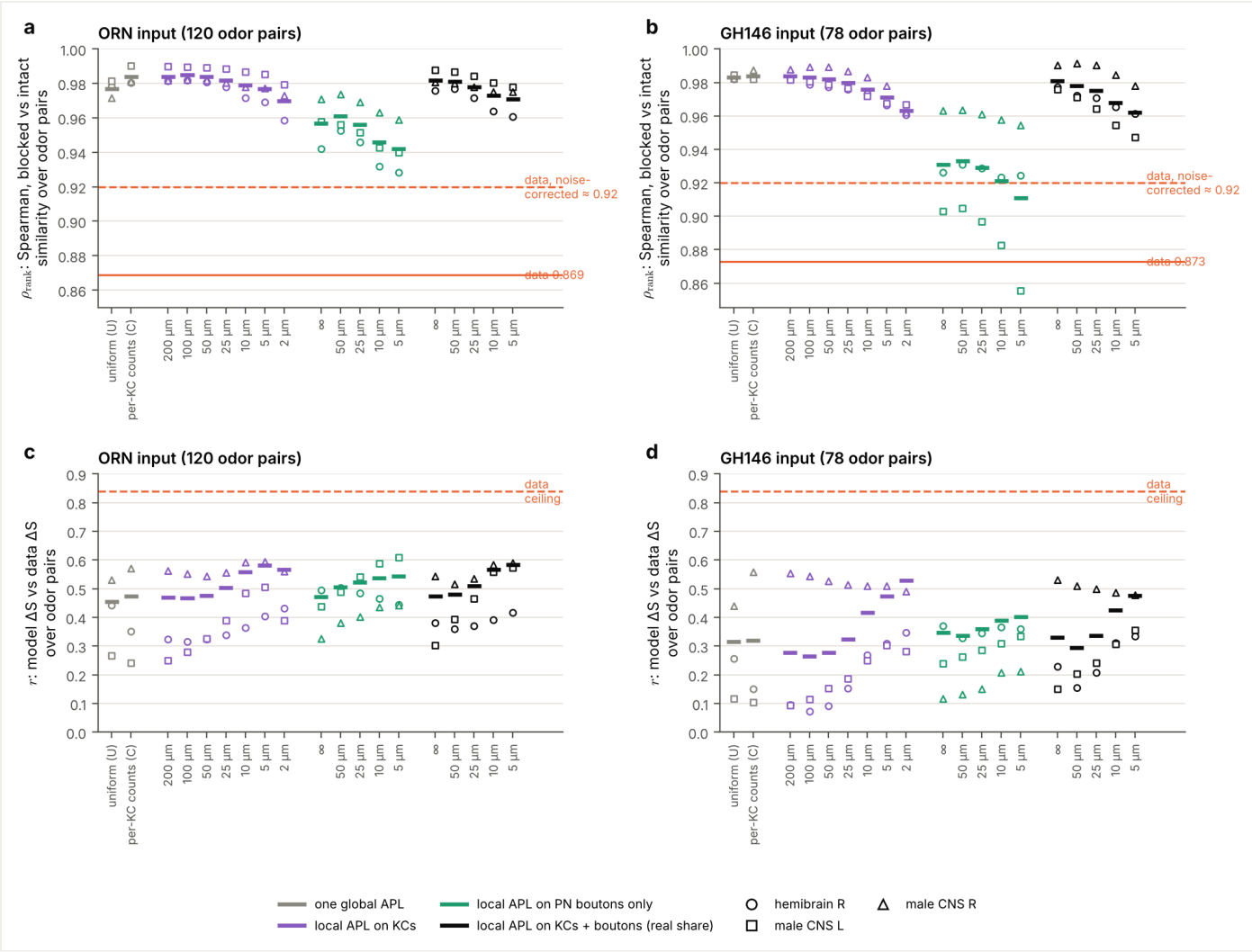


Figure 3. Local inhibition on KCs improves the pattern but barely reorders the pairs. Official ROIs. x-axis: model versions, grouped by color: one global APL (U, C); local APL on KCs (λ from 200 to 2 µm); local APL on PN boutons only (λ = ∞ to 5 µm); local APL on KCs and boutons with boutons at their real share of APL's output. Open markers: the three mushroom bodies; bar: their mean (for **c**–**d** the bar is r of the similarity averaged over mushroom bodies, which is how Report 07 computes it, so it can lie above all three markers). **a**–**b**, rank order ρ_{rank}; lower means inhibition reorders the odor pairs more. Orange lines: data, raw (solid) and corrected for recording noise (dashed). **c**–**d**, pattern r; orange dashed line: the highest r the data's reliability allows (about 0.84). Left: ORN input, 16 odors; right: GH146 input, 13 odors.

Table 2. Main numbers, official ROIs. Selected versions; all are in the summary JSON files. Mean ΔS for the bouton versions is the mean over mushroom bodies of each one's mean change.

Input	Version	ρ_{rank} (mean of MBs)	pattern r	mean ΔS
ORN	uniform (U)	0.977	0.455	0.079
ORN	per-KC counts (C)	0.984	0.475	0.049
ORN	local, KCs, λ = 100 µm	0.985	0.468	0.044
ORN	local, KCs, λ = 25 µm	0.982	0.503	0.044
ORN	local, KCs, λ = 10 µm	0.979	0.558	0.044
ORN	local, KCs, λ = 5 µm	0.977	0.581	0.044
ORN	local, KCs, λ = 2 µm	0.970	0.567	0.045

Input	Version	ρ_{rank} (mean of MBs)	pattern r	mean ΔS
ORN	local, boutons only, λ = ∞	0.957	0.472	0.077
ORN	local, boutons only, λ = 10 µm	0.946	0.538	0.075
ORN	local, boutons only, λ = 5 µm	0.942	0.544	0.077
ORN	local, KCs + boutons, λ = ∞	0.982	0.473	0.054
ORN	local, KCs + boutons, λ = 10 µm	0.973	0.566	0.051
ORN	local, KCs + boutons, λ = 5 µm	0.971	0.584	0.051
ORN	**data** (TNT vs TNT3c)	**0.869** (≈ 0.92 noise-corrected)	ceiling ≈ 0.84	**0.178**
GH146	uniform (U)	0.983	0.316	0.061
GH146	per-KC counts (C)	0.984	0.319	0.037
GH146	local, KCs, λ = 100 µm	0.983	0.265	0.034
GH146	local, KCs, λ = 25 µm	0.980	0.324	0.030
GH146	local, KCs, λ = 10 µm	0.976	0.418	0.029
GH146	local, KCs, λ = 5 µm	0.971	0.473	0.028
GH146	local, KCs, λ = 2 µm	0.963	0.528	0.029
GH146	local, boutons only, λ = ∞	0.931	0.347	0.051
GH146	local, boutons only, λ = 10 µm	0.921	0.389	0.048
GH146	local, boutons only, λ = 5 µm	0.911	0.403	0.051
GH146	local, KCs + boutons, λ = ∞	0.981	0.330	0.040
GH146	local, KCs + boutons, λ = 10 µm	0.968	0.426	0.033
GH146	local, KCs + boutons, λ = 5 µm	0.962	0.477	0.033
GH146	**data** (TNT vs TNT3c)	**0.873** (≈ 0.92 noise-corrected)	ceiling ≈ 0.84	**0.201**

Answer, official ROIs

Rank order: no. A local APL on the KCs leaves ρ_{rank} at 0.970–0.985 (ORN) and 0.963–0.984 (GH146), like U (0.977, 0.983) and far from the data's ≈ 0.92. Only inhibition placed entirely on the PN boutons gets close: 0.942 (ORN), 0.911 (GH146). At the boutons' real share of APL's output (KC + boutons) it is back to 0.971–0.982, 0.962–0.981.

Pattern: somewhat. Short length constants raise r with the data's pattern of changes: best local on KCs 0.581 at 5 µm (ORN), 0.528 at 2 µm (GH146), vs U 0.455 / 0.316 and C 0.475 / 0.319. Shuffling KCs within class lowers r, so the real placement of each KC on APL matters, not only locality: ORN: r 0.581 vs shuffles 0.427 (5–95% 0.28–0.55), p = 0.02; GH146: r 0.528 vs shuffles 0.186 (5–95% −0.07–0.41), p = 0.01.

Size: no. The mean change in similarity of U and the local versions on KCs is 24%–44% of the data's (ORN), 14%–30% of the data's (GH146).

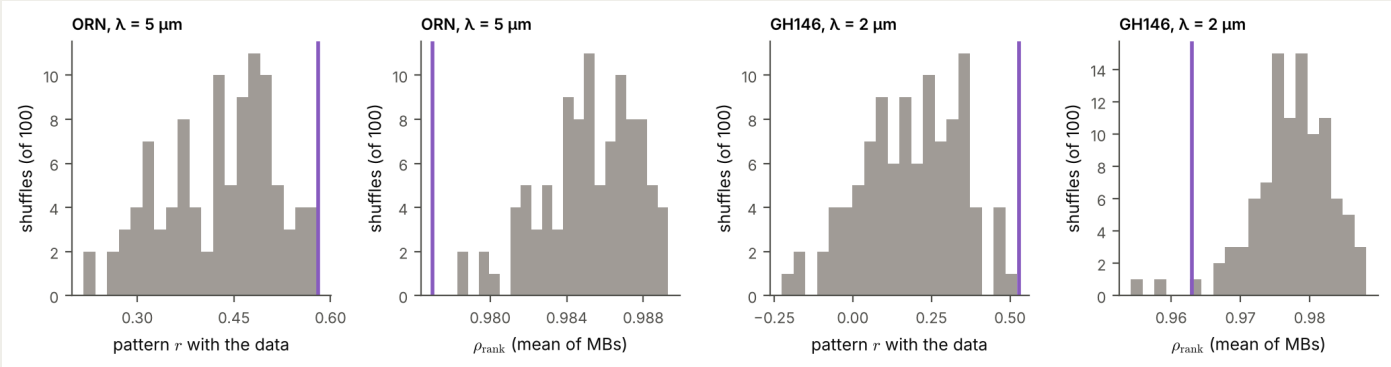


Figure 4. Does the real position of each KC on APL matter? Within-class site shuffles. Gray: 100 shuffles in which KCs swap their APL synapse sites with other KCs of the same class (in all three mushroom bodies at once); purple line: the real wiring. Shown for the local version with the best pattern r for each input table (named in the panel title). Left pair: ORN input; right pair: GH146 input. In each pair, the first panel is the pattern r (higher = closer to the data), the second the rank order (lower = more reordering).

The shuffles show that the improvement in pattern comes partly from where each KC's synapses are, not only from inhibition being local. Rank order with real wiring vs shuffles: ORN 0.977 vs 0.985, p = 0.01; GH146 0.963 vs 0.977, p = 0.03. Any difference is a few thousandths, far less than the gap to the data.

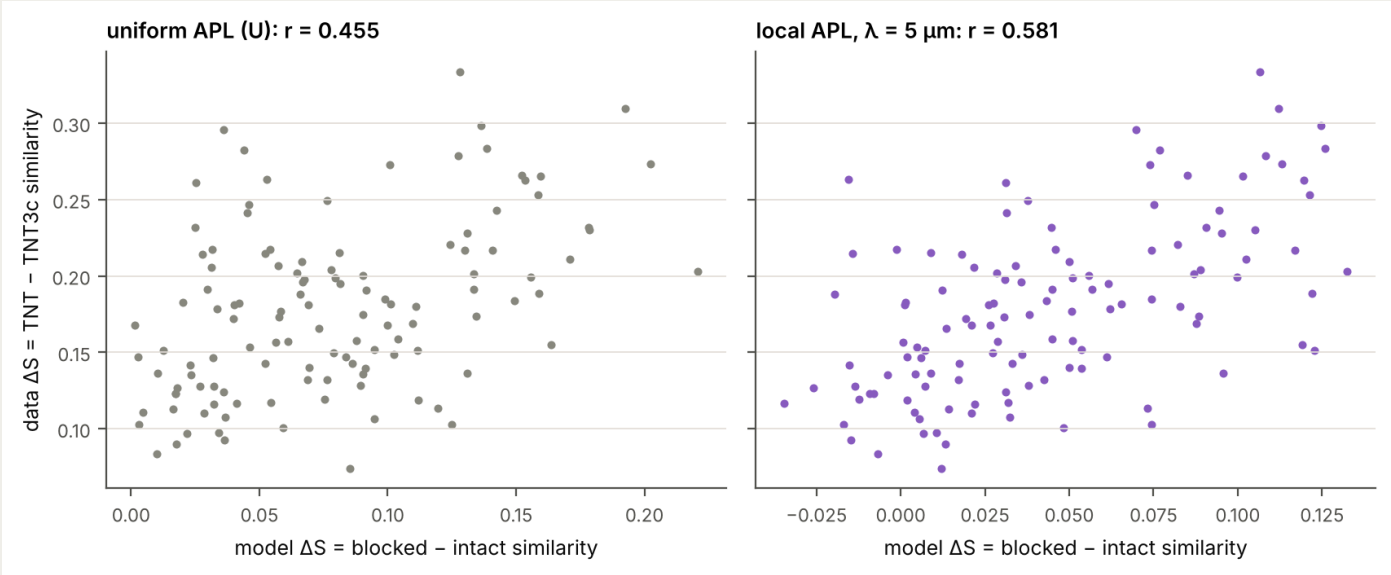


Figure 5. What the pattern r measures, for the uniform and the best local APL. Each dot is one of the 120 odor pairs (ORN input, official ROIs). x: change in cosine similarity of the model KC responses when APL is blocked (similarity averaged over the three mushroom bodies); y: the change in the data (TNT minus TNT3c, averaged over recordings). Left: uniform APL; right: the local version with the best pattern r. The x-axes span different ranges: the model's changes are smaller than the data's.

5 Second version: synapses outside the official ROIs added

The same model was run on the second version of the tables (Report 02), which mostly adds the back of the male CNS calyx: more KC ↔ APL synapses per KC and more APL sites on the skeleton. Figure 6 is Figure 3 for this version, and Table 3 compares the main numbers.

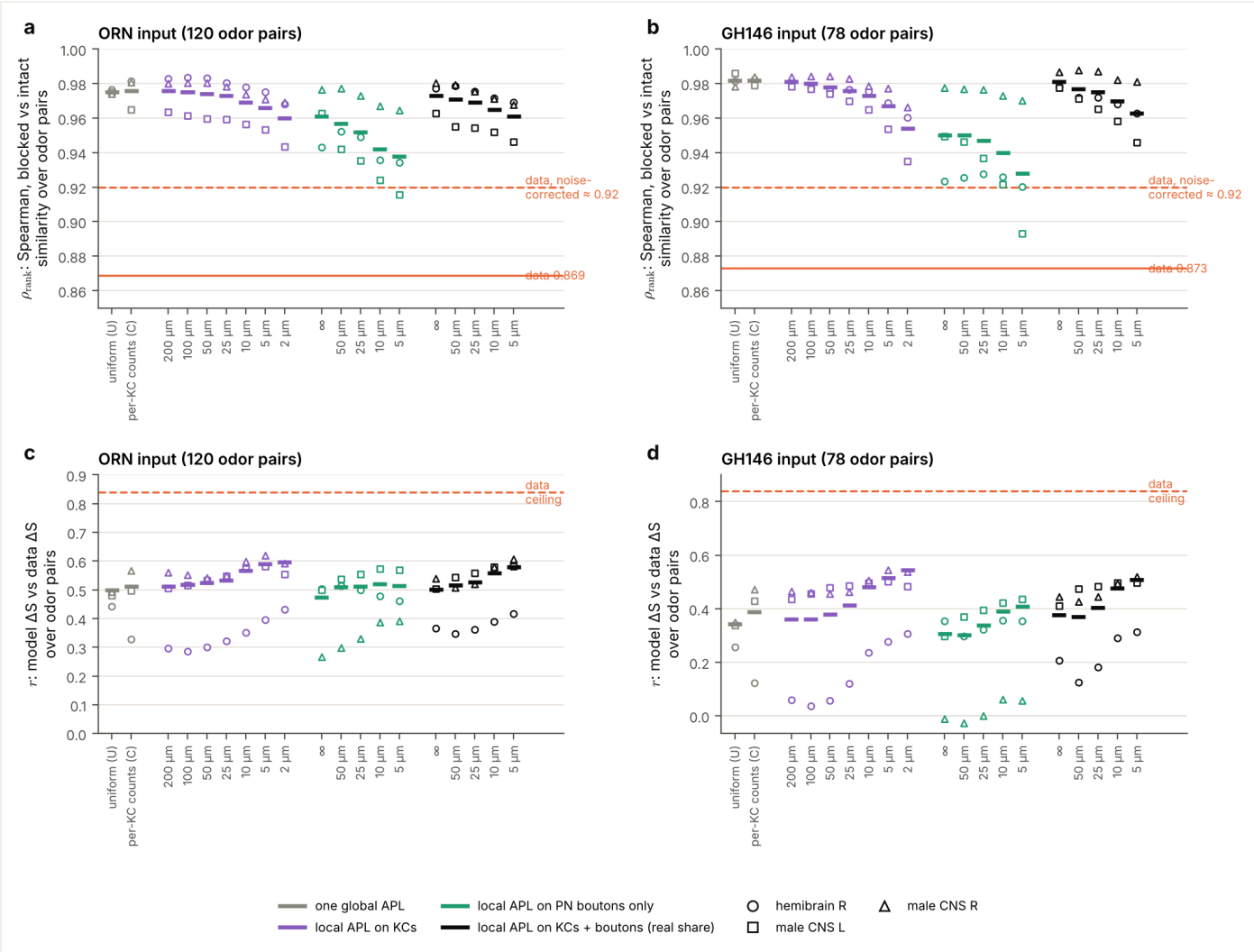


Figure 6. Rank order and pattern, second version. As Figure 3, with the tables of `with_outside_ROI_synapses/`.

Table 3. Official ROIs vs the second version. Means over the three mushroom bodies (ρ_{rank}) and r of the MB-averaged change, as in Figure 3.

Input	Version	ρ_{rank} official	ρ_{rank} second version	r official	r second version
ORN	U	0.977	0.975	0.455	0.500
ORN	C	0.984	0.976	0.475	0.512
ORN	local on KCs, λ = 5 µm	0.977	0.966	0.581	0.591
ORN	local on KCs, λ = 2 µm	0.970	0.960	0.567	0.596
ORN	boutons only, λ = 5 µm	0.942	0.938	0.544	0.515
ORN	KCs + boutons, λ = 5 µm	0.971	0.961	0.584	0.580
GH146	U	0.983	0.982	0.316	0.342
GH146	C	0.984	0.982	0.319	0.388
GH146	local on KCs, λ = 2 µm	0.963	0.954	0.528	0.545
GH146	boutons only, λ = 5 µm	0.911	0.928	0.403	0.409
GH146	KCs + boutons, λ = 5 µm	0.962	0.963	0.477	0.508

The conclusions are the same in both versions: local inhibition on KCs does not lower the rank order much, bouton-only inhibition comes closest, and short length constants raise the pattern r. Shuffle test

(pattern r), second version: ORN r 0.596 vs 0.483, p = 0.16; GH146 r 0.545 vs 0.286, p = 0.01.

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6 What this means, and caveats

- A non-spiking APL whose inhibition stays local predicts **which** odor pairs change most when APL is blocked better than a uniform or count-weighted APL does. Whether the real positions of each KC's synapses add to this beyond locality is mixed: the real wiring beats within-class shuffles for ORN, official ROIs (p = 0.02); GH146, official ROIs (p = 0.01); GH146, second version (p = 0.01), but not for ORN, second version (p = 0.16). This supports the idea that APL's local structure shapes the effect of inhibition on which pairs change.
- It does **not** make inhibition reorder the pairs as much as in the data. In this model, inhibition applied at the KCs, whatever its spatial range, keeps the order of similarities nearly intact (Report 09 shows when that is expected). Inhibition applied before the KCs, on the PN boutons, changes the input itself and reorders more; at the boutons' real synapse share it is too weak to matter here.
- The model's mean changes are 24%–44% (ORN, official), 14%–30% (GH146, official), 33%–48% (ORN, second version), 15%–28% (GH146, second version) of the data's, so something not in the model (e.g. stronger or nonlinear inhibition, KC-to-KC or other interactions, recording noise that differs between genotypes) also contributes.

Caveats

λ = 2 µm is close to the skeleton's resolution (synapses snap 0.4–0.8 µm to nodes), so the shortest λ values are approximate.

The best λ was picked before the shuffle test; the p values do not correct for that choice.

Synapse counts stand in for synaptic strength, and the passive decay $e^{-d/\lambda}$ ignores branch thickness and the signal's time course (a steady state is assumed).

A threshold-linear network with strong feedback can have more than one steady state; the solver finds the one reached from the blocked responses.

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